

## 4.2 Fact 2: Though overall employment continues to grow, employment growth for young workers in particular has been stagnant

Figure 4 shows overall employment trends across age groups, pooling together all occupations. Overall employment remains robust, coinciding with a low national unemployment rate in the post-pandemic period. However, Figure 4 suggests some leveling off in employment growth for young workers relative to other age groups, consistent with recent discussion of a potentially worsening job market for entry-level workers (Chen, 2025; Federal Reserve Bank of New York, 2025).

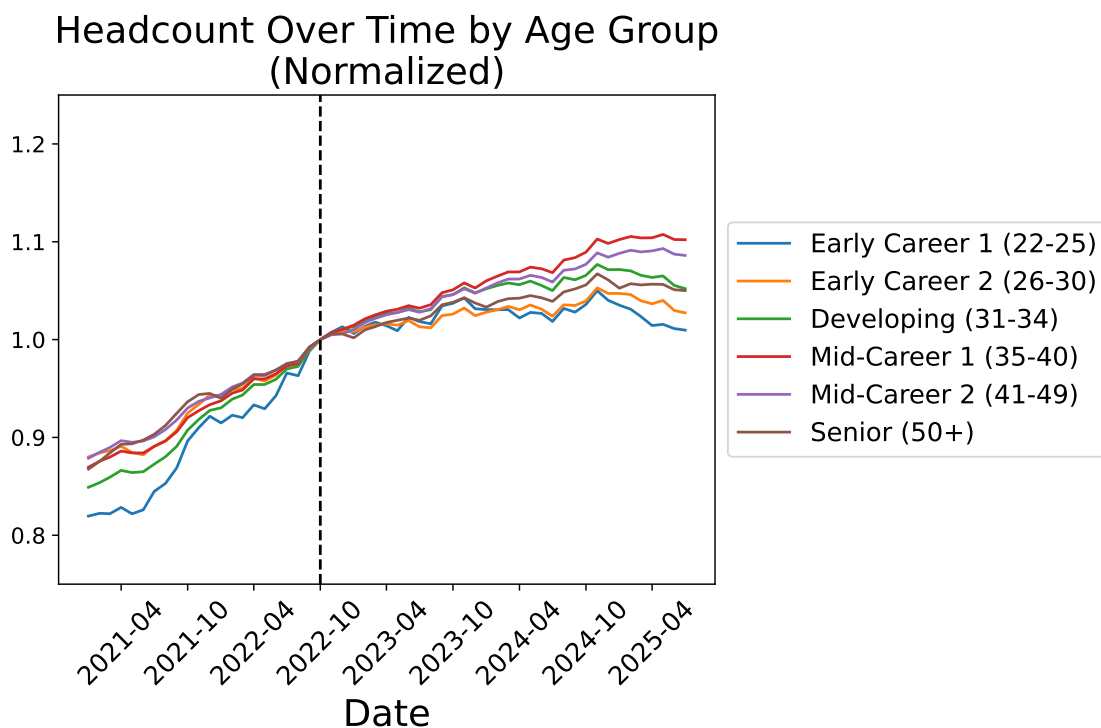


Figure 4: Employment changes by age. Including all occupations.

Figure 5 offers insight into how these trends relate to AI exposure. For each age group, employment growth from late 2022 to July 2025 was 6-13% for the lowest three AI exposure quintiles, with no clear ordering in employment growth by age. In contrast, for the highest two exposure quintiles employment for 22-25 year olds declined by 6% between late 2022 and July 2025, while employment for workers aged 35-49 grew by over 9%. These results show that declining employment in AI-exposed jobs is driving tepid overall employment growth for workers between the ages of 22 and 25.

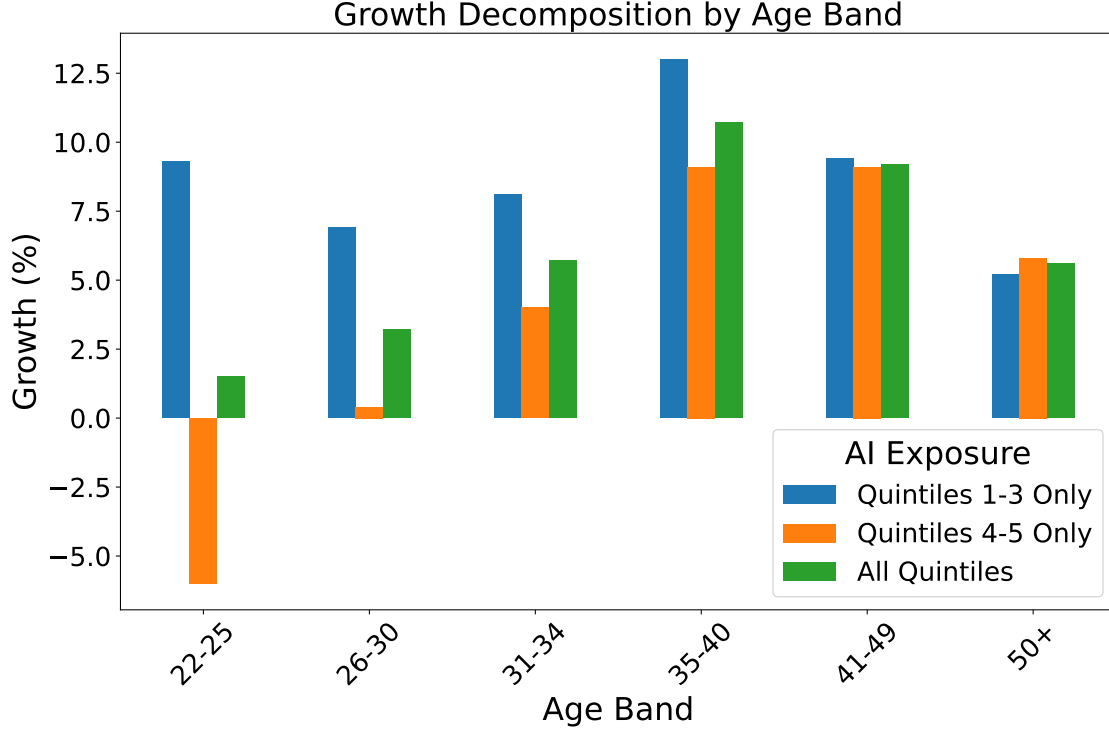


Figure 5: Growth in employment between October 2022 and July 2025 by age and GPT-4  $\beta$ -based AI exposure group.

While these findings suggest divergent employment outcomes by AI exposure for young workers, we caution the trends observed in these first two facts could be driven by other changes in the US economy. Our subsequent facts evaluate the robustness of the results to alternative analyses.

#### 4.3 Fact 3: Entry-level employment has declined in applications of AI that *automate* work, with muted changes for *augmentation*

AI exposure can either complement or substitute for labor. These may have very different implications for the labor market (Brynjolfsson, 2022).

To assess how employment patterns differ based on the complementarity or substitutability of AI with labor, we use data on generative AI usage from the Anthropic Economic Index (Handa et al., 2025). The Index provides an estimate of the share of queries that pertain to each occupation. In addition, for each task it reports estimates of the share of queries pertaining to that task that are “automative,” “augmentative,” or none of the above. We use this information as an estimate of

whether usage of AI for an occupation is primarily a substitute or complement for labor.<sup>15</sup> Table A1 shows example occupations that are in the highest and lowest exposure category for each measure.

Figure 6 shows employment changes by overall prevalence of related Claude queries. The patterns match the findings using the Eloundou et al. (2024) measures closely. Figure 7 likewise shows that the occupations with the highest estimated automation shares have experienced declining employment for the youngest workers.

In contrast, Figure 8 indicates that the occupations with the highest estimated augmentation shares have *not* experienced a similar pattern. Employment changes for young workers are not ordered by augmentation exposure, as the fifth quintile has among the fastest employment growth. The findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.<sup>16</sup>

#### 4.4 Fact 4: Employment declines for young, AI-exposed workers remain after conditioning on firm-time shocks

While our results so far are consistent with the hypothesis that generative AI is causing a decline in entry-level employment, there are plausible alternative explanations. One class of explanations is that our patterns are explained by industry- or firm-level shocks correlated with sorting patterns by age and measured AI exposure. For example, one possibility is that young workers with high measured AI exposure are disproportionately likely to sort to firms heavily susceptible to interest rate increases.

We test for a class of such confounders by controlling for a rich set of fixed effects. For each age group, we estimate the Poisson regression

$$\log(E[y_{f,q,t}]) = \sum_{q' \neq 1} \sum_{j \neq -1} \gamma_{q',j} 1\{t = j\} 1\{q' = q\} + \alpha_{f,q} + \beta_{f,t} + \epsilon_{f,q,t} \quad (4.1)$$

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<sup>15</sup>Figures A2 and A3 show that automation and augmentation results are similar when dropping occupations with low overall Claude usage.

<sup>16</sup>Note that occupations in the first two quintiles of the augmentation measure have very low Claude usage overall, with the average occupation in these quintiles comprising 0.01% and 0.09% of conversations, respectively. These occupations have a high share of conversations that are classified as neither automative nor augmentative. In contrast, occupations in the third through fifth quintiles average 0.47%, 0.39%, and 0.33% of Claude conversations. For the automation measure, overall Claude usage increases on average with the automation share, with the lowest exposure group averaging 0.05% of conversations and the highest group averaging 0.73%.

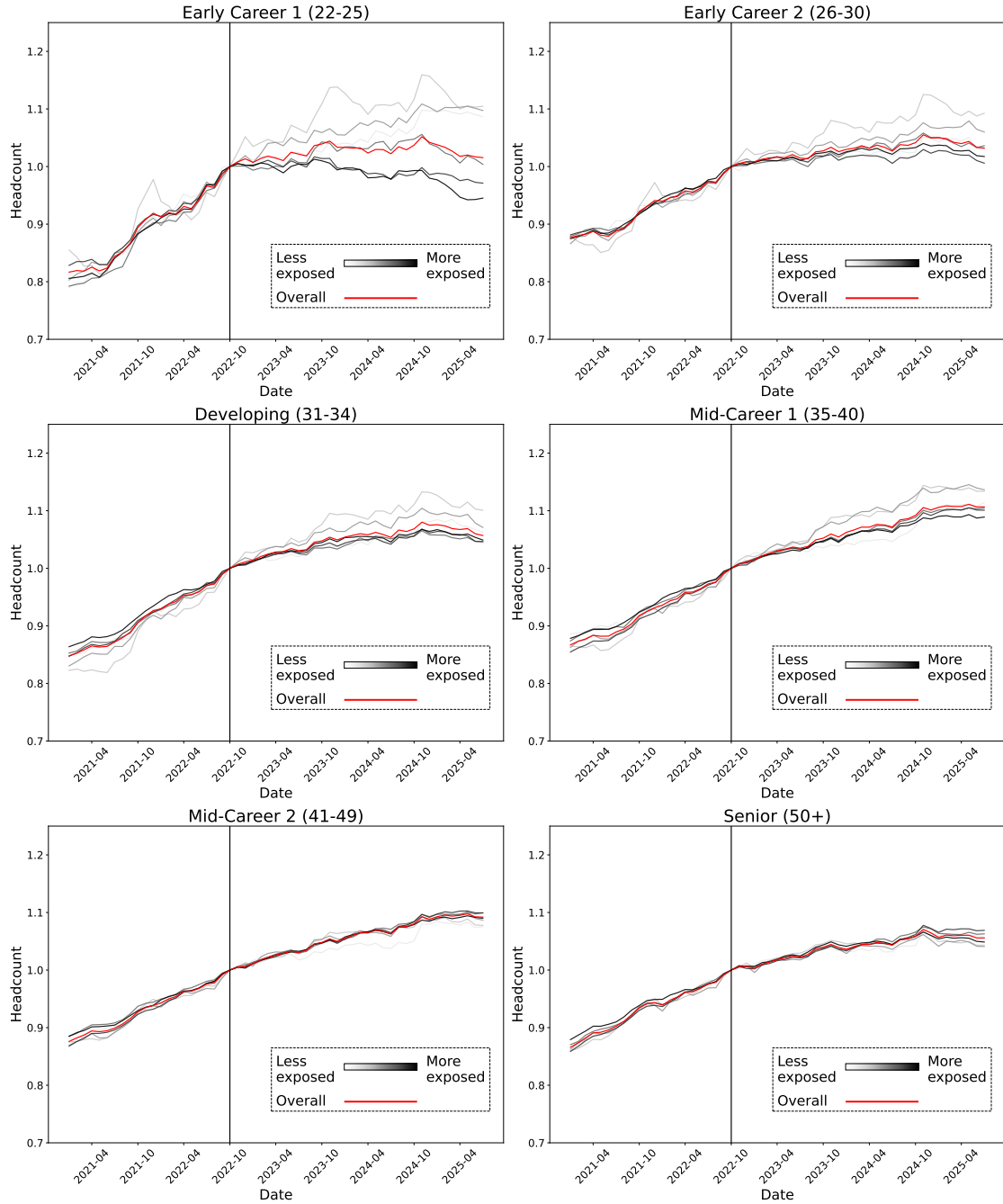


Figure 6: *Overall Claude usage.* Employment changes by age and exposure quintile using Claude usage data from [Handa et al. \(2025\)](#). Exposure quintiles are defined based on the share of queries to Claude that relate to tasks associated with an occupation. Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles. Occupations whose associated tasks all have fewer than the minimum number of queries to appear in the usage data are treated as a separate category, coded as 0.